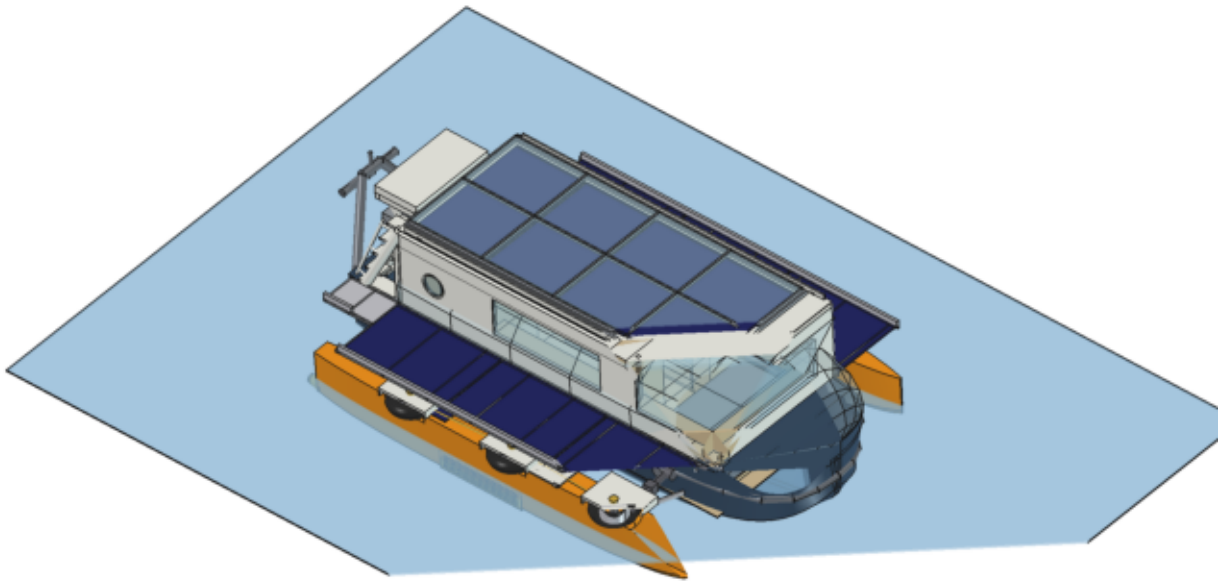


BOAT-HOME

Road-Towable Solar Trimaran

design study · 2026



7.2 m · 2.5 m beam · 2 535 mm on the road

4.40 kWp solar · 50 kWh battery · 4.8 kn maximum

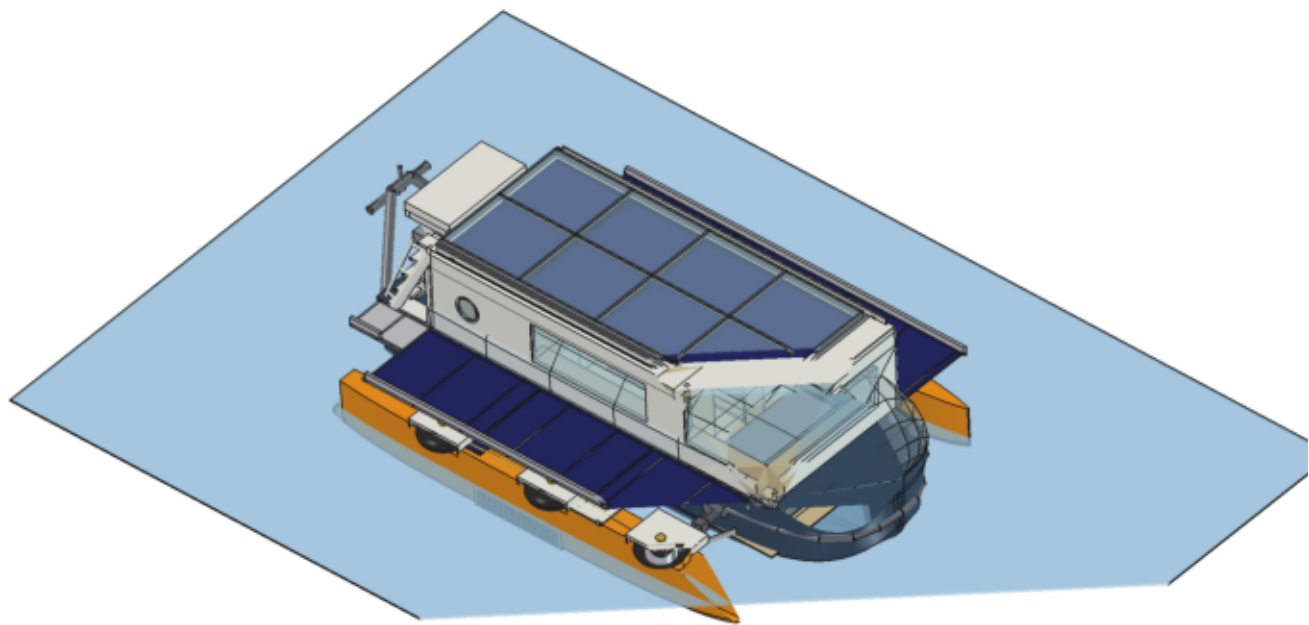
5 berths · 1 850 mm headroom · 12.1 m² floor

its own trailer: six driven wheels inside the floats

Boat-Home — Road-Towable Solar Trimaran

A 7.2 m Dutch-barge-style boat-home that is its own trailer: the stabiliser floats fold underneath the hull and carry six driven wheels, so one Mercedes-Viano-class car tows it on German roads and it drives itself in and out of the water on a slipway. Solar-electric, weed-proof waterjets, and a walk-on glass sun deck — you stand on the glass, the panels live safely underneath it.

Parametric CAD in FreeCAD (Python-scripted); every legal, structural and hydrostatic constraint is asserted in code, so a dimension cannot drift without the build failing.



cruise

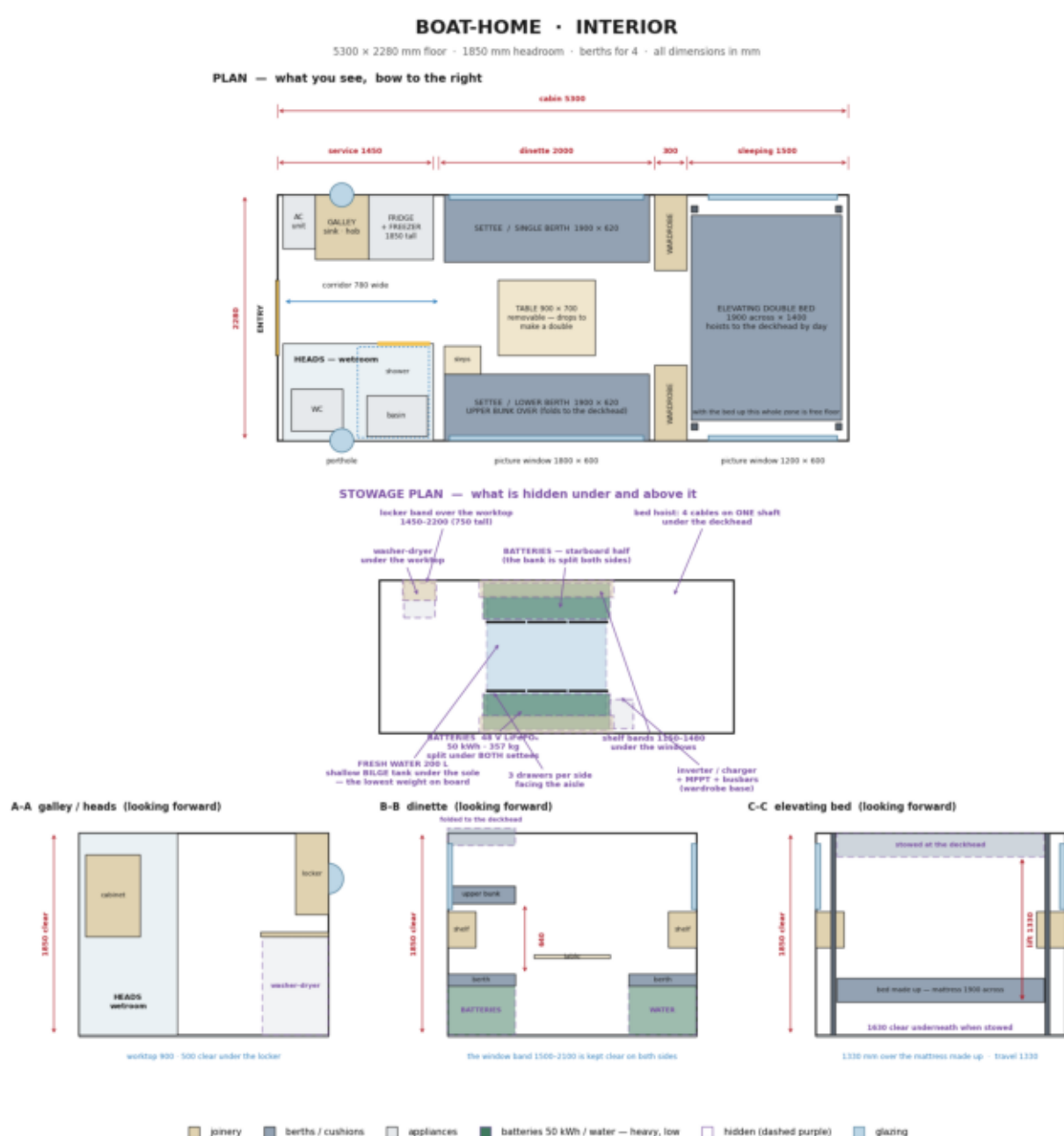
1. Principal dimensions

Length over hull	7 200 mm
Length with drawbar out	9 094 mm
Beam, hull	2 500 mm
Beam, road (folded)	2 535 mm (StVZO limit 2 550)
Beam, afloat (floats out)	4 716 mm
Height on the road	3 009 mm (limit 4 000)
Air draft afloat	2 267 mm
Draft	260 mm light, ~300 mm loaded
Displacement	1 970 kg at the 260 mm waterline
Design all-up mass	2 000 kg (see §7 — this is the tight one)
Ground clearance, road	482 mm

Track / wheelbase	2 270 / 3 400 mm
Coupling height / tongue load	445 mm / +100 kg

Living quarters

Cabin	5 300 × 2 280 mm inside
Headroom	1 850 mm clear over the sole
Floor area	12.1 m ²
Berths	5 — athwartships double, bunk (2), single settee, dinette double
Heads	1 400 × 900 wetroom with shower
Galley	induction hob, sink, washer-dryer, full-height fridge/freezer
Fresh water	200 L in a bilge tank under the sole
Glazing	2 picture windows per side, 1 800 × 600 and 1 200 × 600



interior

2. Performance

Installed power: 3×2 kW flush-intake waterjets = 6 kW. Estimates from an ITTC friction line plus a residuary factor for a blunt barge hull; wetted surface 27.3 m² in trimaran trim, 2 600 kg all-up, waterjet propulsive efficiency 0.45. Method in docs/performance.md.

Speed	Shaft power	Range on 45 kWh usable	Endurance
2.0 kn	0.18 kW	500 NM	250 h
3.0 kn	0.56 kW	240 NM	80 h
3.5 kn	1.00 kW	157 NM	45 h
4.0 kn	2.09 kW	86 NM	21 h
4.5 kn	4.24 kW	48 NM	11 h
4.8 kn max	6.0 kW	33 NM	7 h

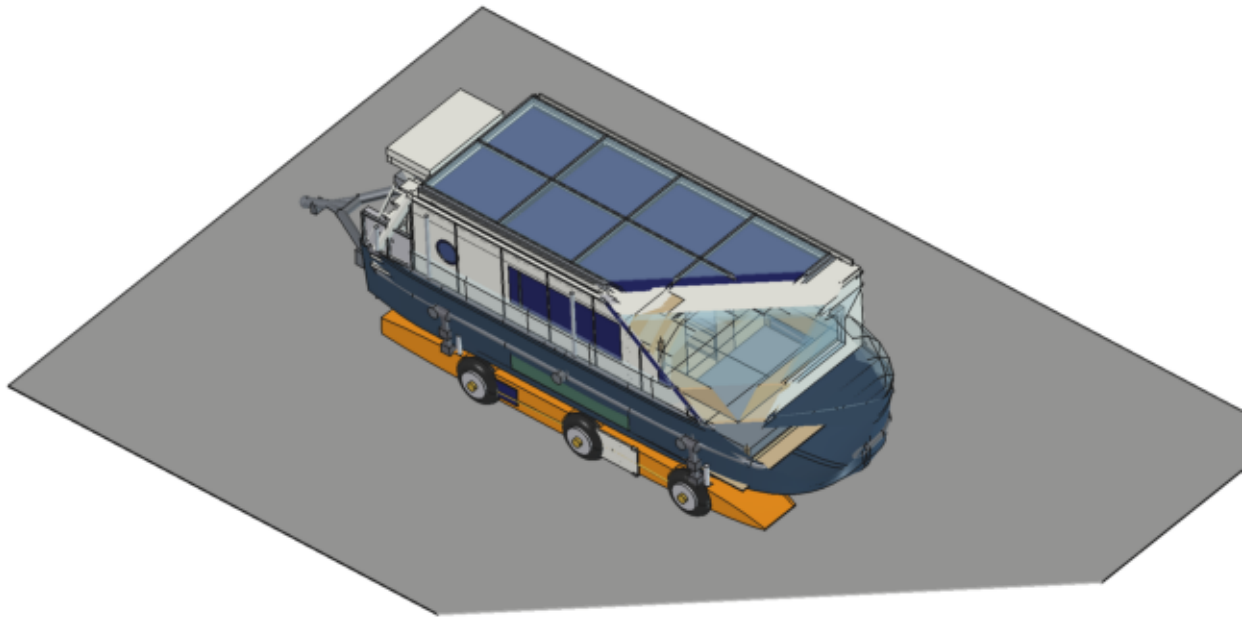
Maximum $\approx$ 4.8 knots, power-limited rather than hull-speed-limited (theoretical hull speed is 6.2 kn). The last 0.3 kn costs as much power as the first 4 — this is a boat to cruise at 3–4 kn, not to push.

Solar-neutral cruising. With 4.40 kWp and $\approx$ 24 kWh on a good summer day, eight hours at 4.2 knots is solar-neutral: the boat moves all day and finishes with a fuller battery than it started. In spring or autumn (12 kWh) that falls to 3.6 kn, and on an overcast winter day (6 kWh) to 2.7 kn.

Battery	48 V LiFePO ₄ , 50 kWh, 357 kg, split under both settees
Solar	4.40 kWp — 4 $\times$ 500 W under the walk-on glass, 6 $\times$ 400 W bifacial on the balconies (4.35 kWp effective)
House load	$\approx$ 2.5 kWh/day (fridge, lights, water, AC standby)
Steering	differential thrust, no rudder
On land	$\leq$ 10 km/h under its own wheels; towed at road speeds

3. The five configurations

Mode	What it is	Renders
road	floats folded under the hull, balconies up over the windows, towed stern-first	iso · side · bow · aft
launch	driving itself down the slipway on six wheels	iso · side · bow · aft
harbor	jack-up: floats carry the boat, keel awash, tyres as rolling quay fenders	iso · side · bow · aft
cruise	trimaran, floats out, balconies down	iso · side · bow · aft
anchor	lying to the stern anchor	iso · side · bow · aft



road

BOAT-HOME • GENERAL ARRANGEMENT • all dimensions in mm

PROFILE — afloat (cruise), bow to the right

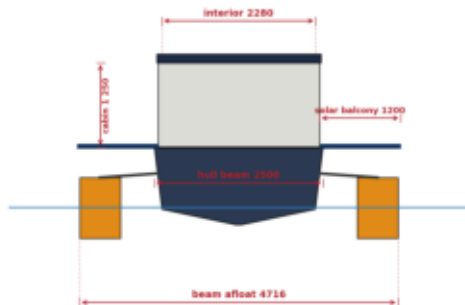


PROFILE — on the road (floats folded flat under the hull, towed stern-first)



6 x 205/70 R15 all-terrain - track 2270 - tongue load +100 kg - road width 2525 (limit 2 550)

MIDSHIP SECTION — afloat (looking forward)



KEY DIMENSIONS

OVERALL

length, hull
length with drawbar
beam - road / afloat
height on the road
air draft afloat
draft / displacement

7 200 mm
9 094 mm
2 525 / 4 716 mm
3 009 mm (StVZO limit 4 000)
2 267 mm
260 mm / +1 970 kg

LIVING QUARTERS

cabin length
cabin width, inside
standing headroom
roof sun deck
floor area
entry door, clear
cockpit floor / bulwark

5 300 mm
2 280 mm
1 850 mm
walk-on glass, 6 panes
= 12.1 m²
700 x 1 700 mm
400 / 750 mm

DECKS & SOLAR

solar balcony
aft passage width
roof terrace
panels

1 200 wide: 0 walkway + 1160 panels
540 mm (part of the folding deck)
5 300 x 2 360 mm
5 laminates + 6 std modules (= 3.1 kWp)

ROAD GEAR

wheels
track / wheelbase
ground clearance
coupling height / tongue
float, each

6 x 205/70 R15 all-terrain
2 270 / 3 400 mm
482 mm
445 mm / +100 kg
6 200 x 600 x 900 mm

general arrangement

4. Key systems

Hangar / running gear — one rigid welded arm per station, a single shoulder pin, 90° swing. Road: float on its side flush under the hull, wheels vertical. Water: float flat, wheels dry above the waterline. Six 205/70 R15 all-terrain wheels, electric-over-hydraulic drive with the machinery sealed inside the floats. → docs/wheels.md

Jack-up stance — the floats hold 3.1 t of buoyancy against a 2.0 t boat, so folding the arms in deep water lifts the hull: the floats end 65 % submerged carrying everything and the keel rides awash. Pontoon GM ≈ 3.1 m.

Propulsion — flush perforated intake grids on the float sides below the waterline (0.5 m/s face velocity, 14 mm holes, so weed drifts past), internal duct to an enclosed pump, tail nozzle. Nothing rotating is reachable by weed. → docs/propulsion.md

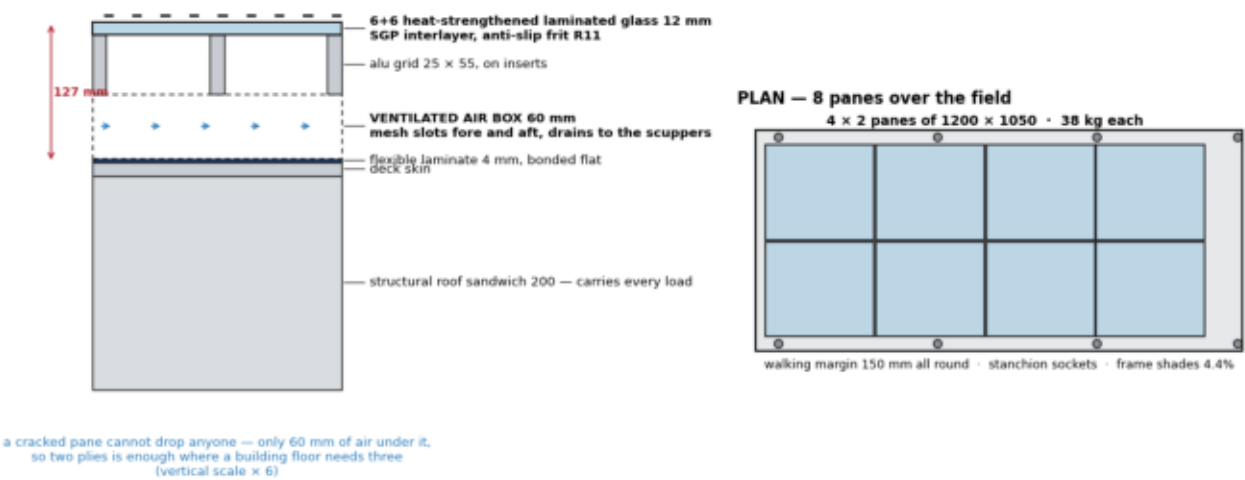
Exoskeleton — an external steel ladder-loop frame carries every point load (arms, balconies, tow, fenders); nothing structural crosses the living volume. → docs/structure.md

Roof deck — walk-on 6+6 laminated glass over a ventilated 60 mm air box on an aluminium grid, solar modules underneath, no moving parts.

Sized for full building-code deck loads: 2 kN/m² plus 2 kN on a 50 × 50 mm patch. → docs/roof.md

ROOF DECK BUILD-UP • walk on the glass, never on the panels

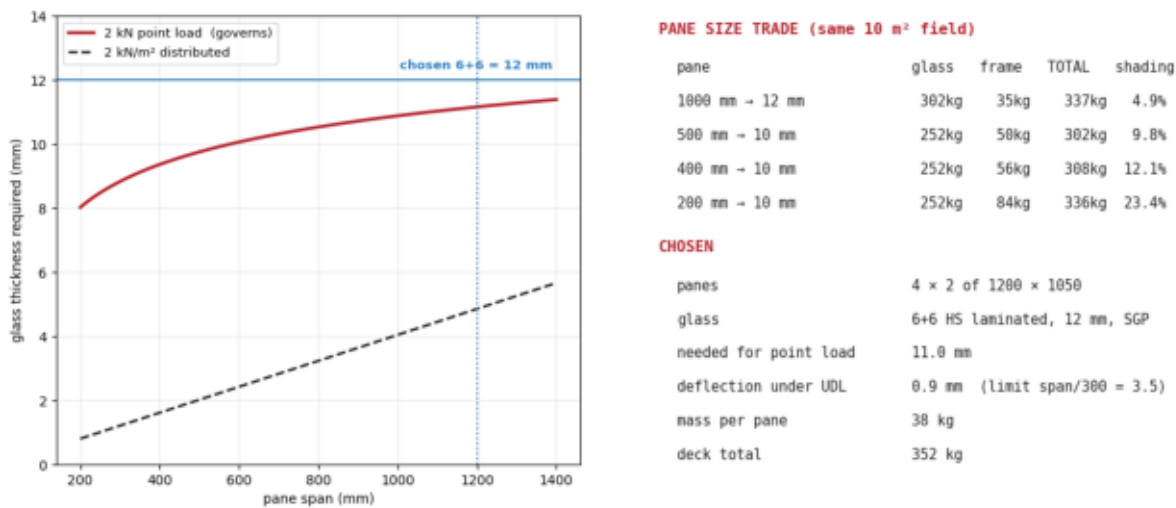
no moving parts anywhere on the roof; the air box ventilates the cells so they run cooler than bonded laminates



roof deck

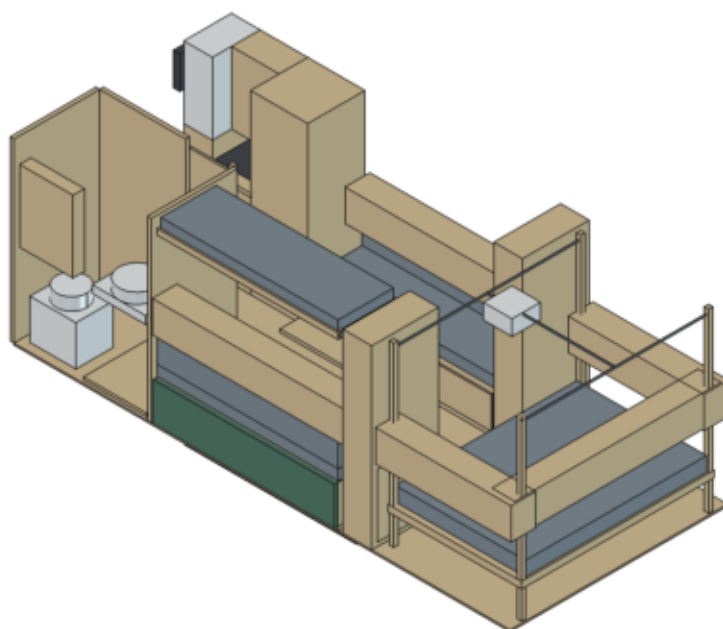
WALK-ON GLASS SIZING • why the pane size hardly matters

the concentrated load governs, and its stress grows with the LOG of the span — not the square of it



glass sizing

Interior — heads, galley, dinette that sleeps two, elevating double bed forward, fold-down bunk to port; batteries and water low and amidships. → docs/interior.md



interior iso

Tow and stern gear — a wide A-arch on the transom pin-locks into a sea gantry or an extensible drawbar; the 2 t electric self-recovery winch and the anchor share the centreline so a ramp pull has no yaw bias.

5. Cost estimate (2026, EUR, materials)

Item	Low	High	Source
Hull shell, structure, fairing	12 000	18 000	est.
Floats, 2 × 6.2 m	3 500	4 500	est.
Exoskeleton frame + tow arch	2 600	3 200	structure.md
Wheels, hubs, in-float drive	5 600	6 500	wheels.md
Propulsion, 3 × 2 kW waterjets	8 800	13 300	propulsion.md
Walk-on glass roof deck	5 110	6 000	roof.md
Solar balconies	1 700	2 000	roof.md
Battery bank, 50 kWh LiFePO ₄	9 000	11 000	at 180–220 €/kWh
Electrics: inverter/charger, MPPT, switchgear, cabling	3 000	4 500	est.
Glazing: winter garden + windows	6 000	9 000	est.
Interior fit-out incl. appliances, heads, AC	12 000	16 000	est.
Paint, antifoul, deck finish	2 000	3 000	est.
Approvals: CE self-certification, TÜV/StVZO trailer	2 000	4 000	est.
Materials total	73 000	101 000	

Built professionally, add labour: ≈ 2 550 h at 60–80 €/h = 150 000–205 000, so a yard-built boat lands near 230 000–300 000. The design assumes a self-build, where that labour

is your own time.

6. Build time

Phase	Hours
Hull shell, bulkheads, fairing	700
Floats, arms, wheels, hangar kinematics	350
Propulsion install and ducting	150
Roof deck, glass, solar, wiring	200
Interior joinery and fit-out	500
Electrics, plumbing, systems	250
Paint and finishing	250
Commissioning, trials, approvals	150
Total	≈ 2 550 h

- Self-build at 15 h/week — ≈ 3.3 years
- Self-build at 25 h/week — ≈ 2 years
- Professional yard, 2–3 people — ≈ 6–9 months elapsed

7. Open risks

1. Mass budget — the binding constraint. The interior alone is 832 kg (the 50 kWh bank is 357 of it) and the glass deck 460 kg, against a 2 000 kg design figure for towing. Hull, frame, floats, wheels, arms and jets do not fit in what is left. Levers, cheapest first: travel with the water tank empty (–200 kg), carry half the bank as removable modules (–180 kg), or re-budget at ~2 500 kg and tow with a heavier car. → interior.md §8

2. TÜV approval of the integrated running gear as a trailer is unprecedented; get an engineer's sign-off early.

3. Jet hydrodynamics are first-order estimates — the intake grids want CFD or a tank test before committing.

4. Arm and shoulder-pin fatigue — each arm carries ~1 000 kg through one pin, cycled every launch.

5. No guardrail on the sun deck — a deliberate choice for looks; that deck is 2.4 m above the water.

8. Documents

Document	Covers
performance.md	speed, range, the resistance model
wheels.md	hangar kinematics, in-float drive, BOM
propulsion.md	waterjets, weed strategy, BOM
structure.md	exoskeleton frame, tow arch, stern gear
roof.md	walk-on glass deck, glass sizing, balconies
interior.md	layout, stowage, services, mass budget
aft_entry.md	companionway, porch, ladder, AC, lockers
glossary.md	vocabulary for every part and term

9. Repository layout

freecad/	the model – the source of truth
params.py	ALL dimensions, kinematics, checks() asserts
build_boat.py	geometry builders -> boat_<mode>.FCStd
view.sh	build + open in the FreeCAD GUI
capture.py	wipes shots/, re-renders every configuration
interior_shots.py	interior renders
ga_drawing.py	dimensioned general-arrangement sheet
interior_plan.py	interior sheet: plan + stowage plan + sections
roof_cards.py	roof-deck spec cards
shots/	current renders (regenerated, never stale)
docs/	
*.md	per-system design studies (table above)
images/	dimensioned spec cards and drawings
make_pdf.py	renders this README to docs/boat-home.pdf
drafts/	original concept PDF + STL sketch
*.scad, Makefile	legacy OpenSCAD sketch (superseded)

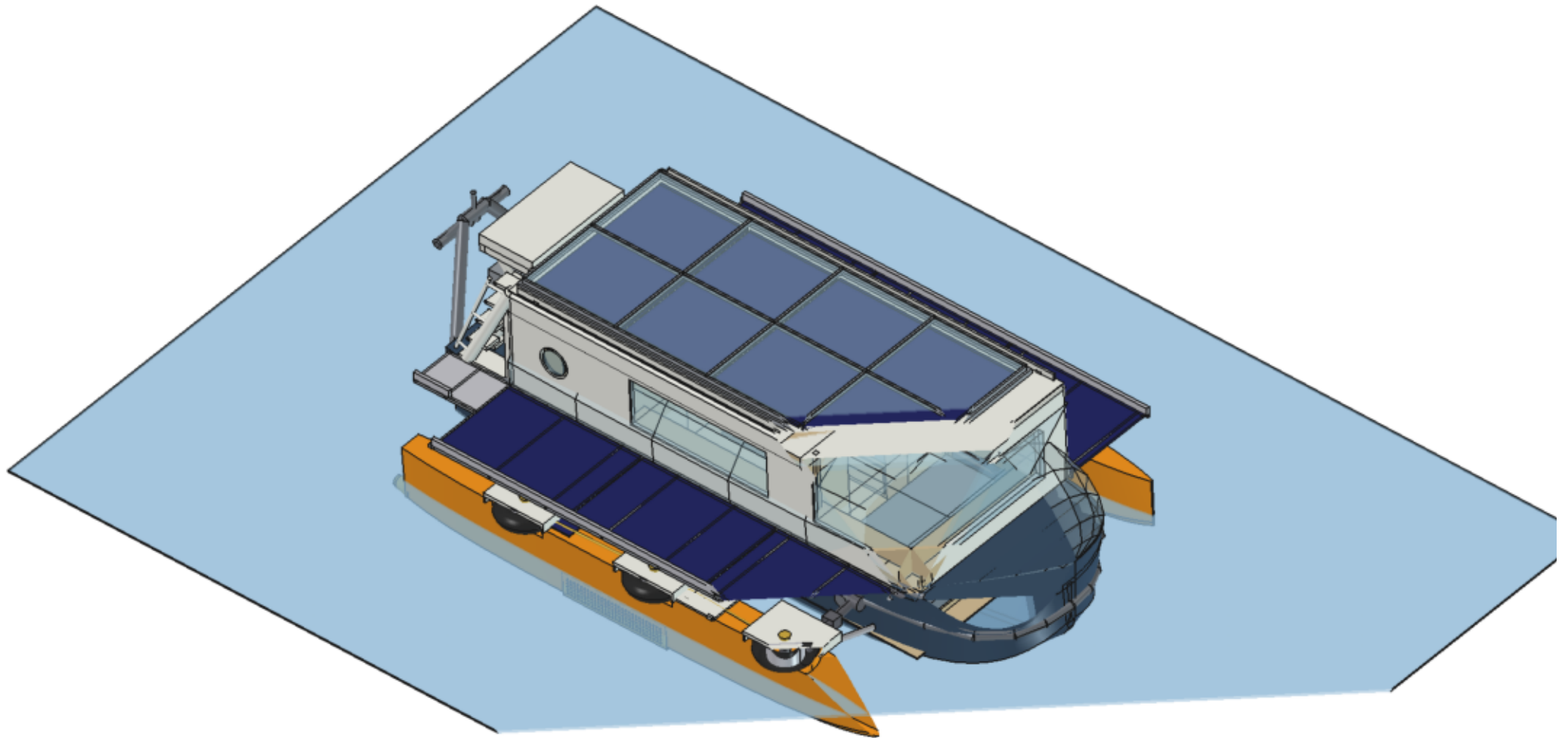
10. Working on it

```
./freecad/view.sh                # open in the GUI
~/bin/FreeCAD.AppImage freecad/capture.py    # regenerate all renders
~/bin/FreeCAD.AppImage freecad/interior_shots.py
python3 freecad/ga_drawing.py          # arrangement sheet
python3 freecad/interior_plan.py        # interior sheet
python3 freecad/roof_cards.py           # roof spec cards
python3 docs/make_pdf.py                # this README as a PDF
python3 -c "import sys; sys.path.insert(0,'freecad'); import params; params.checks()"
```

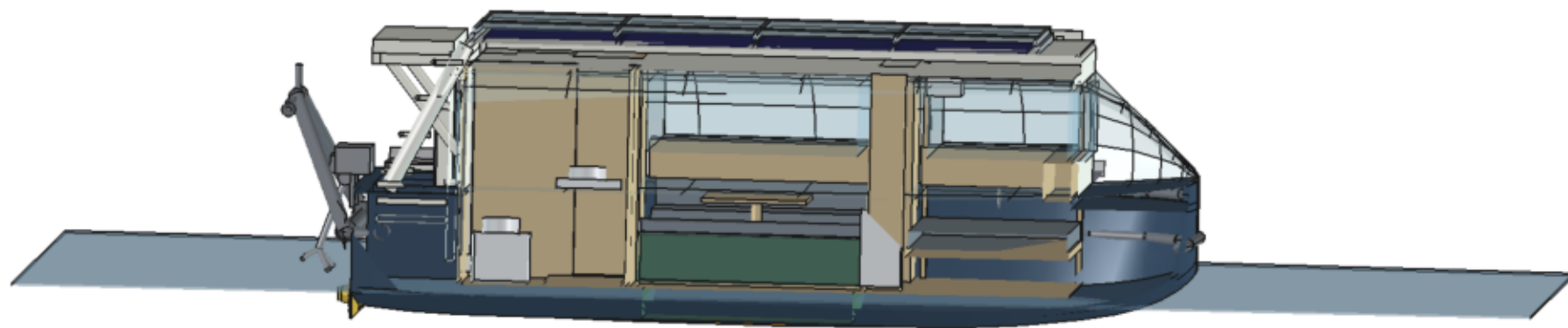
params.py is the single source of truth. checks() recomputes road legality (StVZO), submergence margins, stability, glass thickness, panel fit and interior clearances from the same numbers the geometry uses — change a dimension and the asserts tell you what broke.

DRAWINGS AND RENDERS

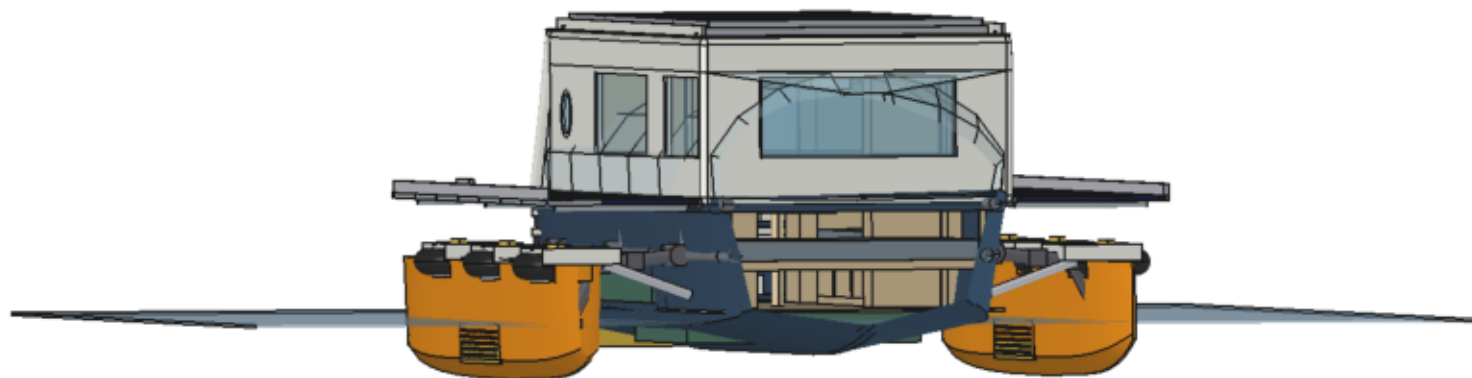
30 plates



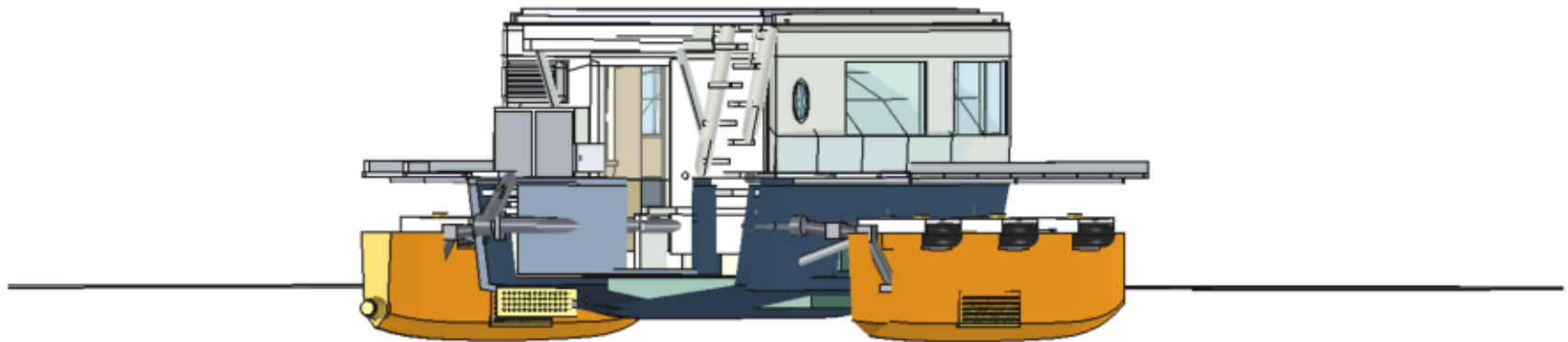
cruise — trimaran, floats out



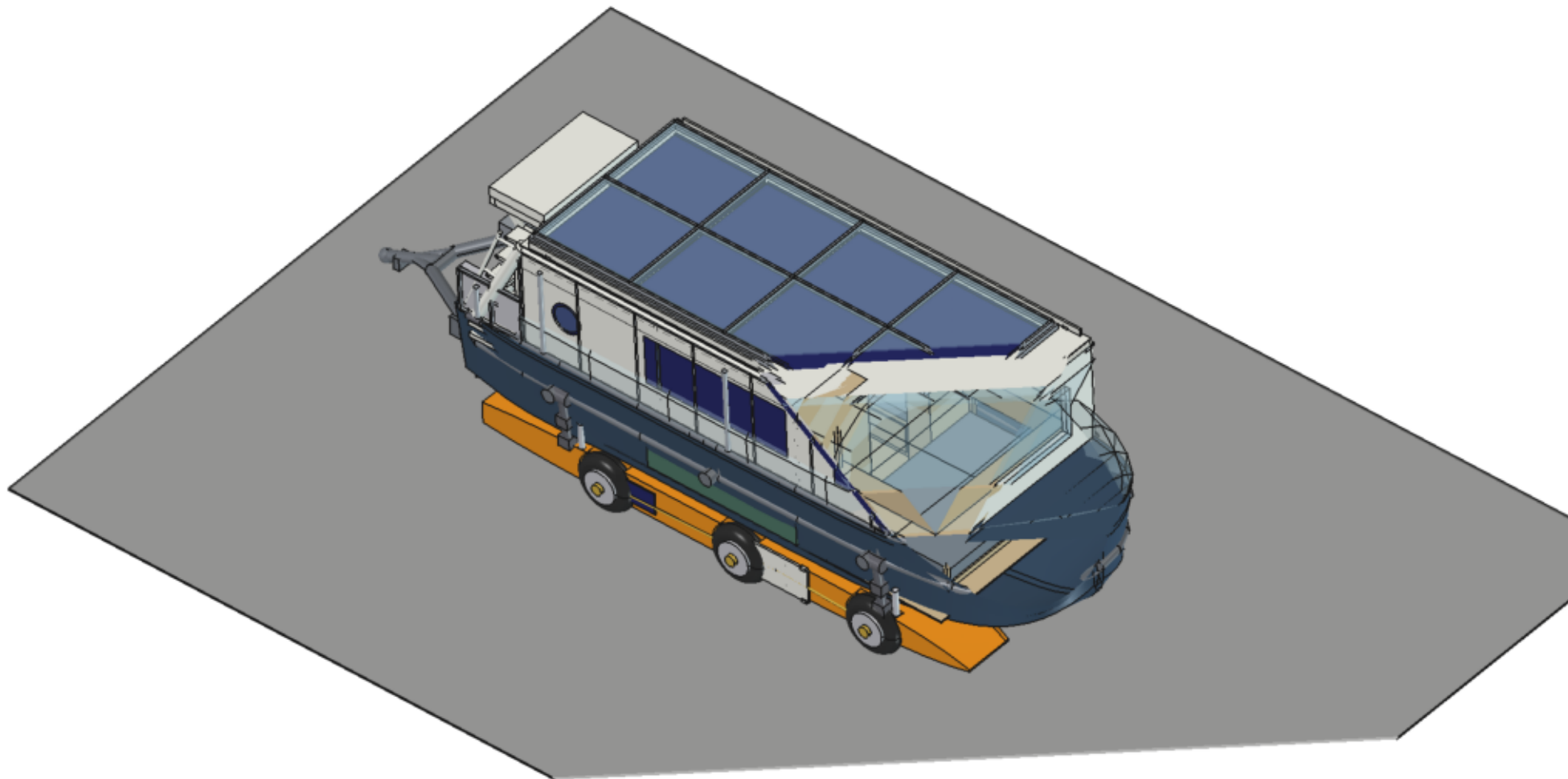
cruise — side



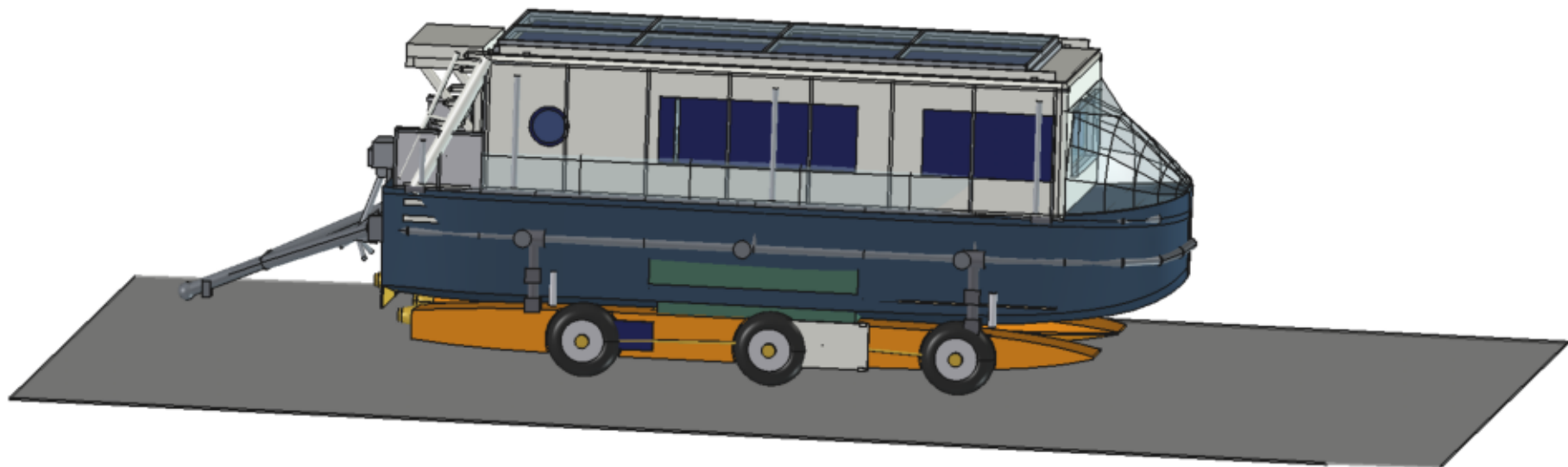
cruise — bow



cruise — stern



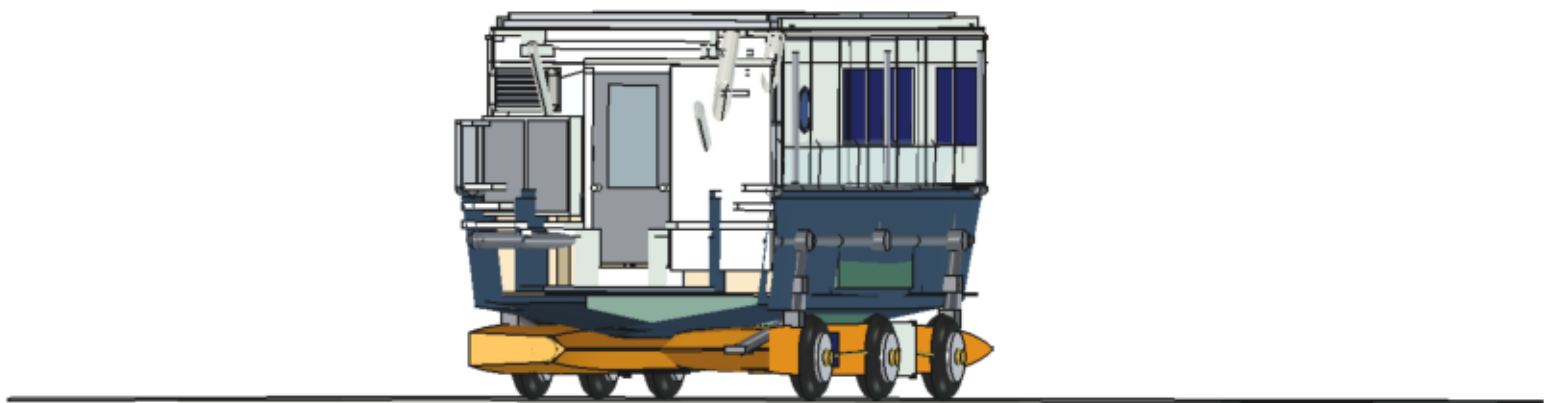
road — folded, towed stern-first



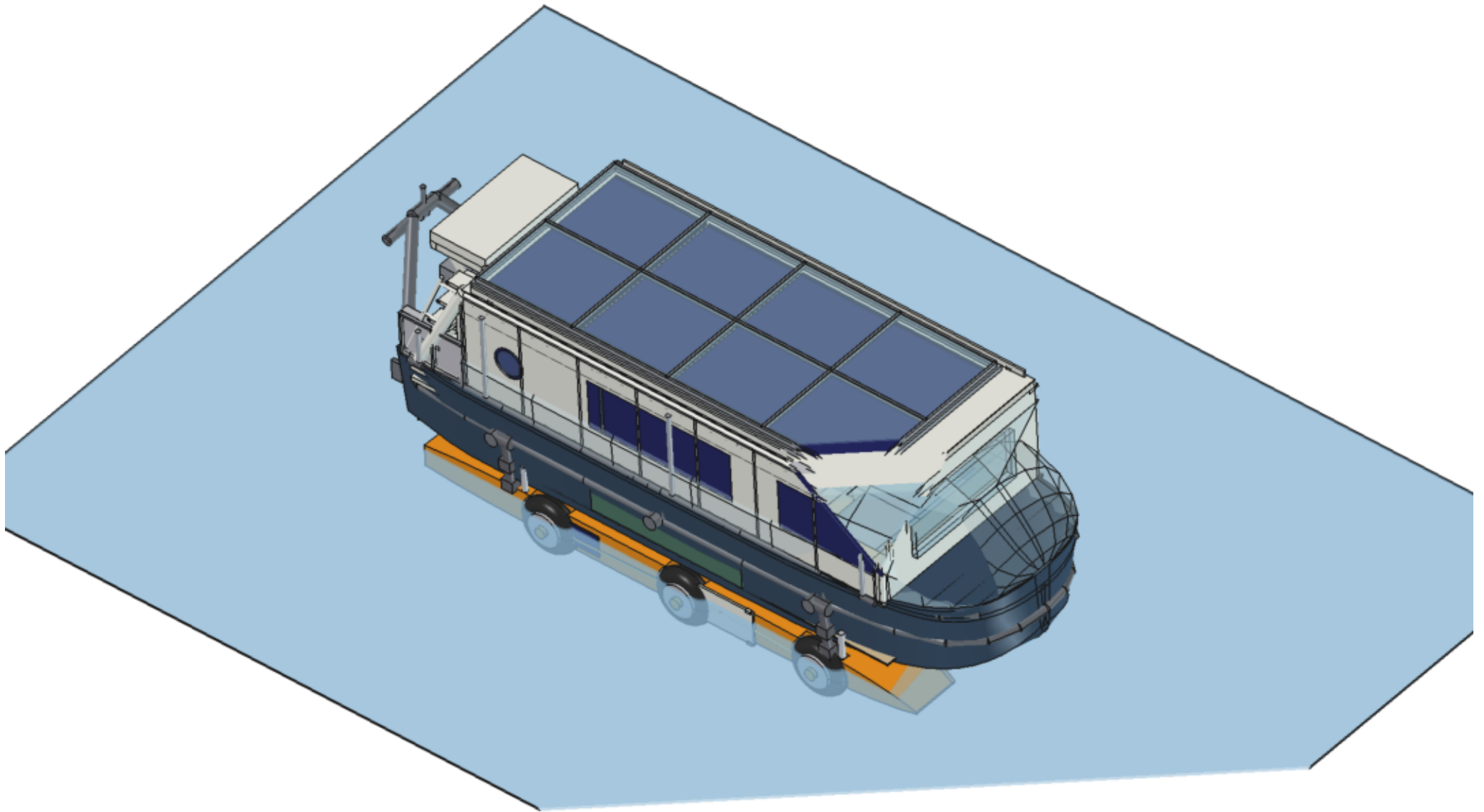
road — side



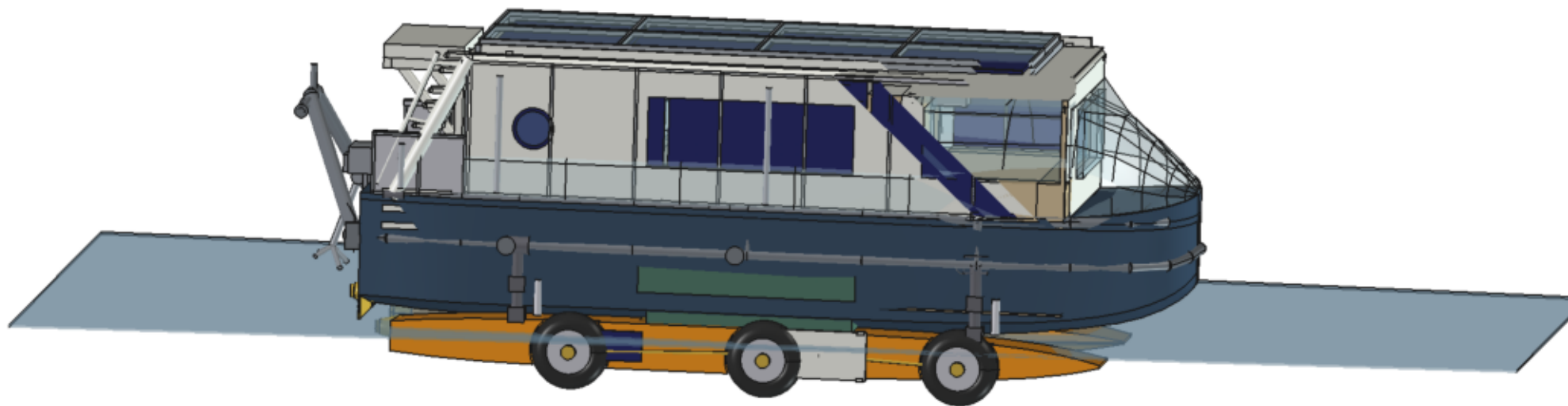
road — bow



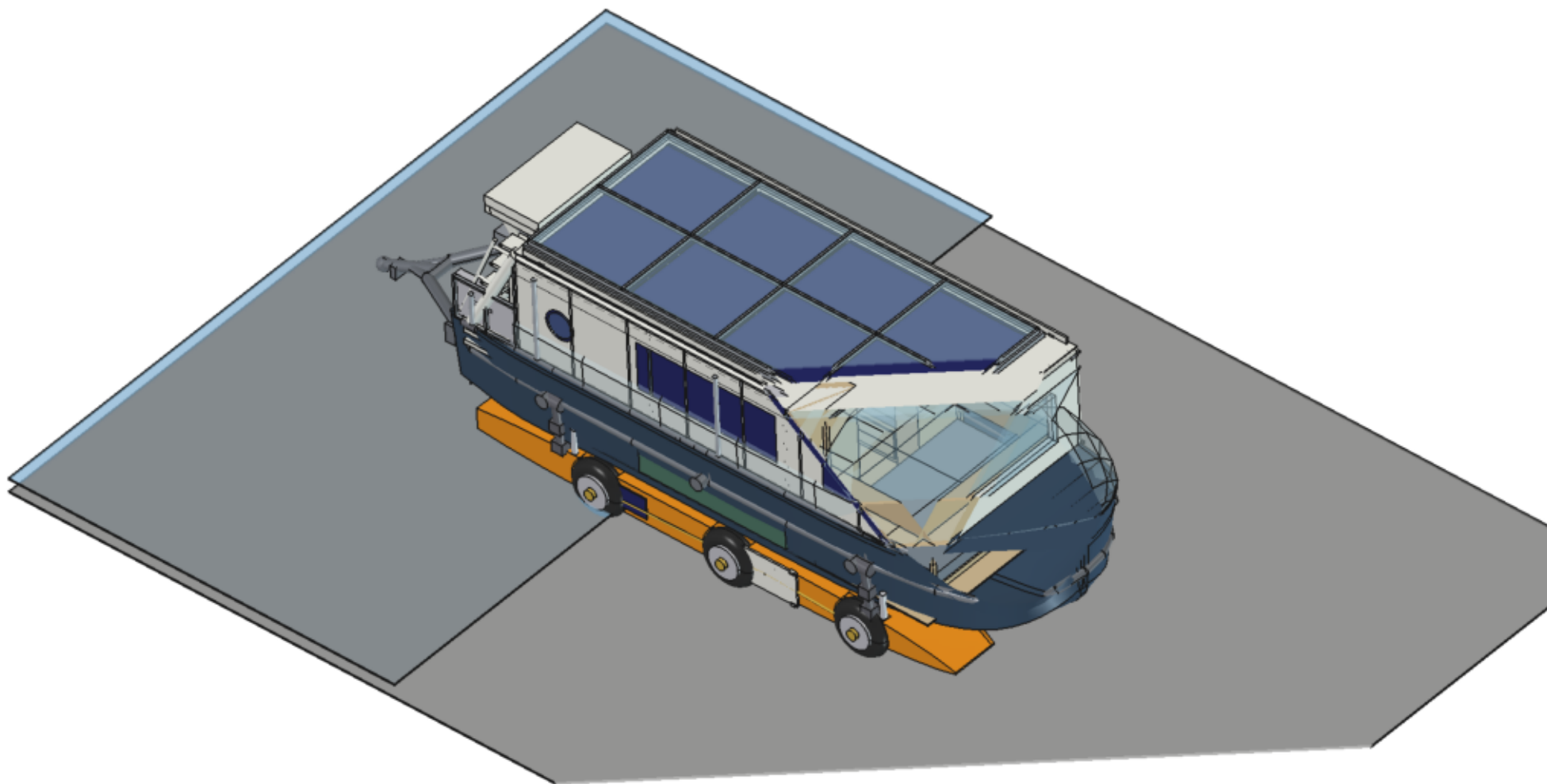
road — stern



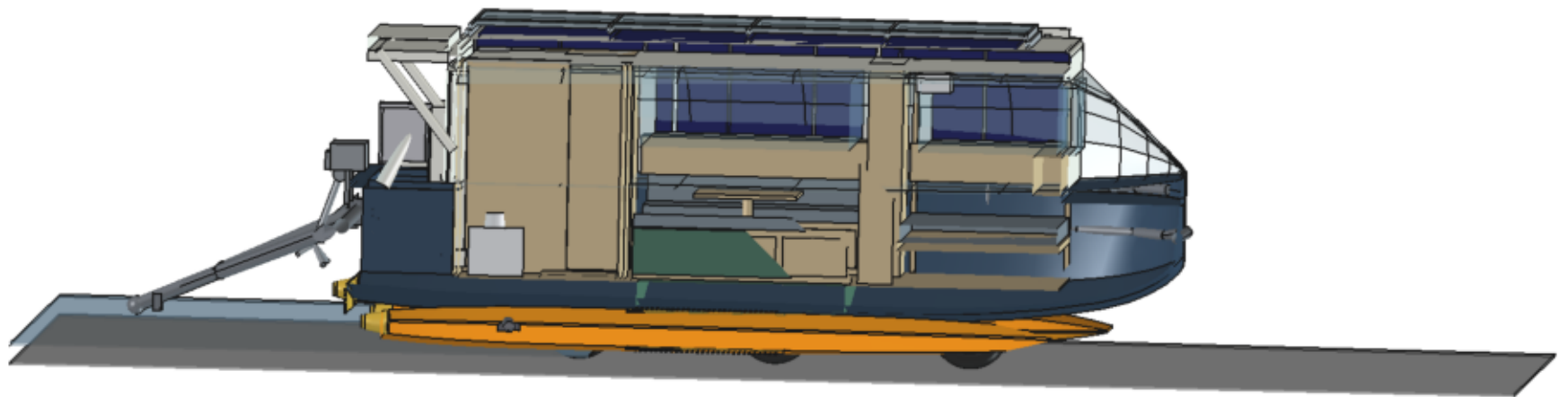
harbour — jack-up stance



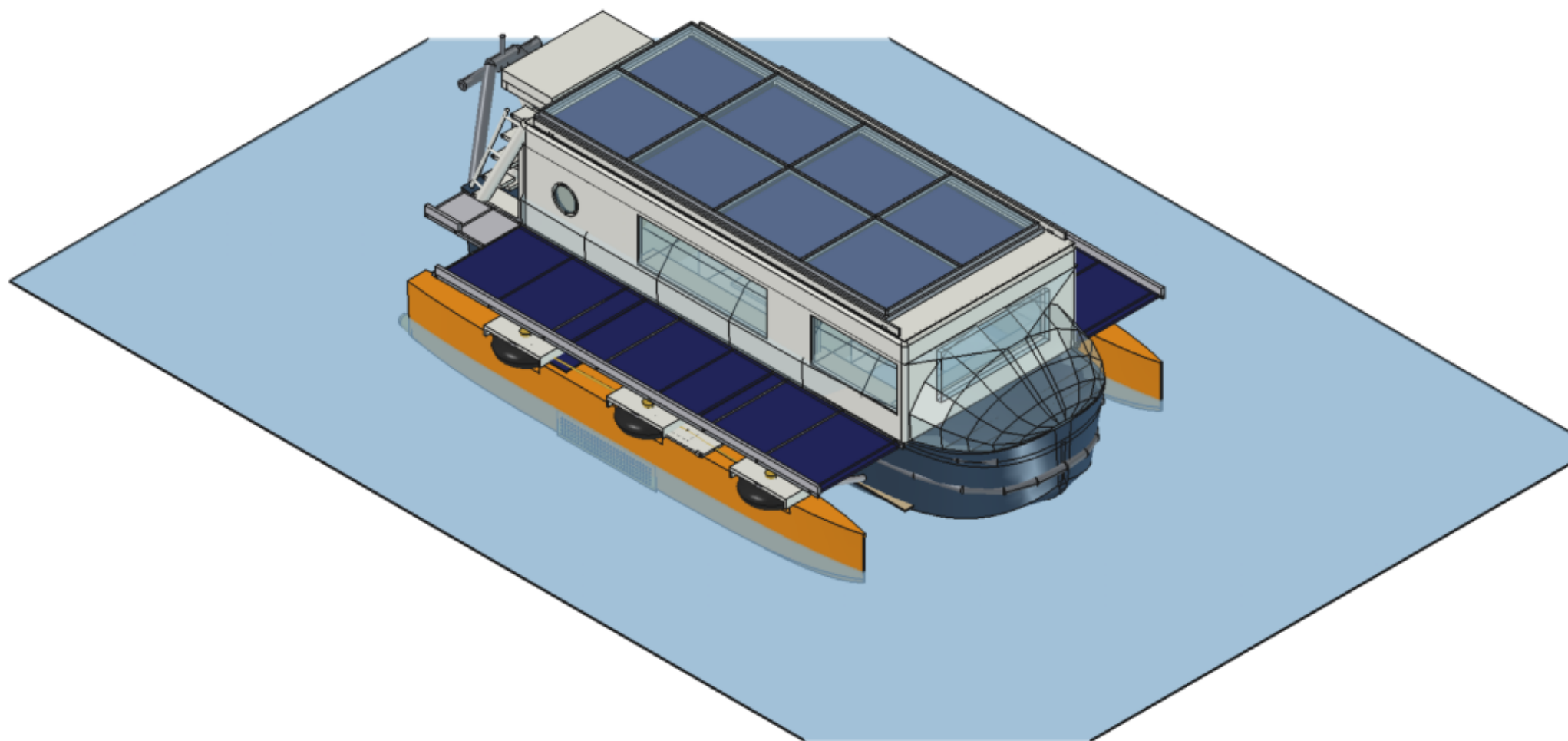
harbour — side



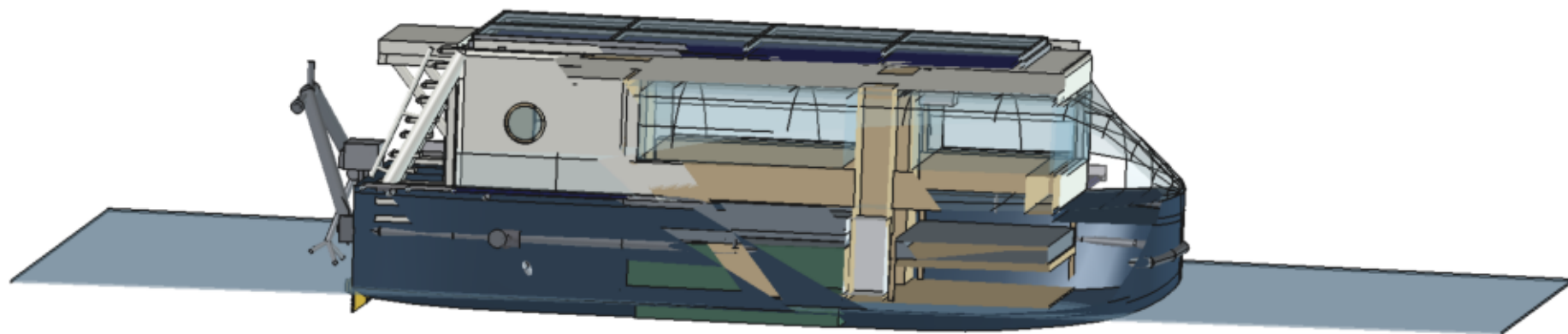
launch — driving down the slipway



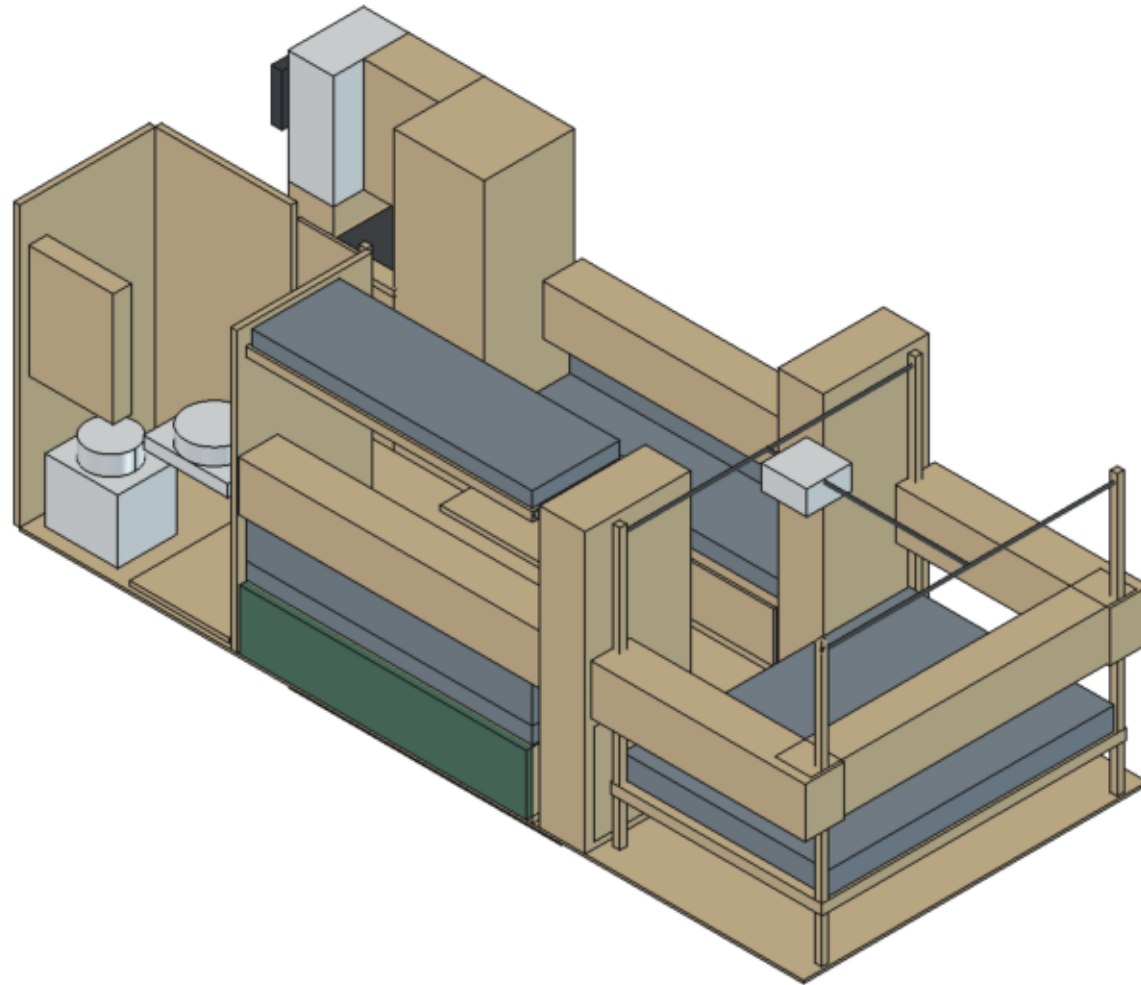
launch — side



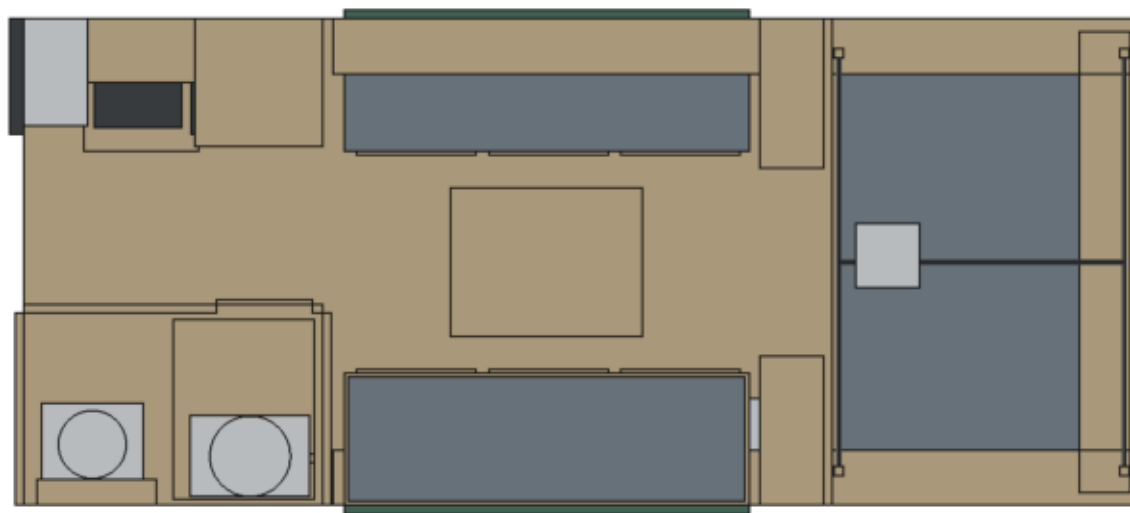
anchor



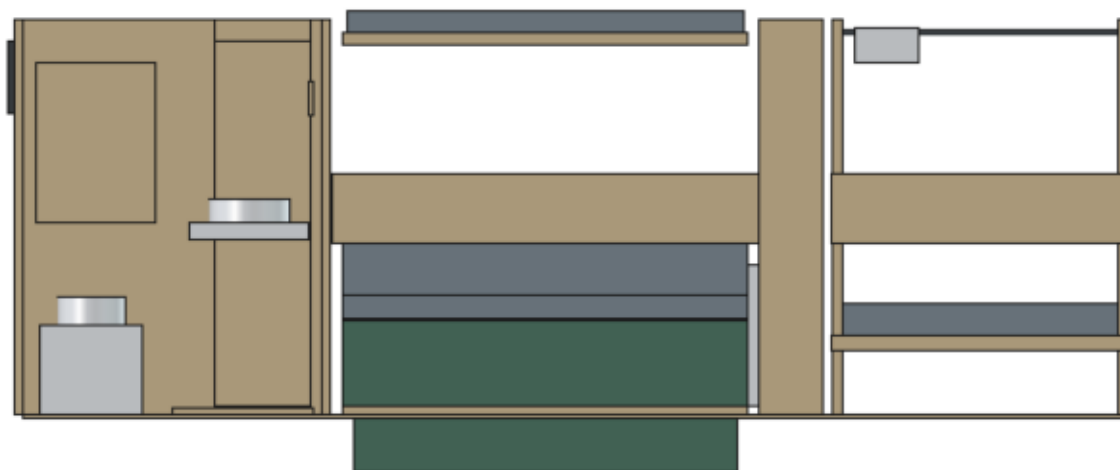
anchor — side



interior — fit-out



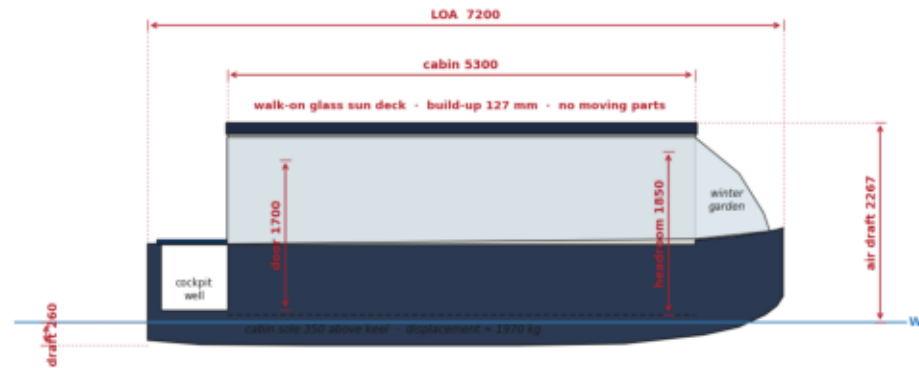
interior — plan



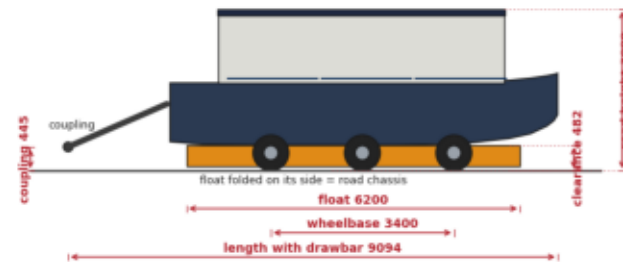
interior — side elevation

BOAT-HOME · GENERAL ARRANGEMENT · all dimensions in mm

PROFILE — afloat (cruise), bow to the right

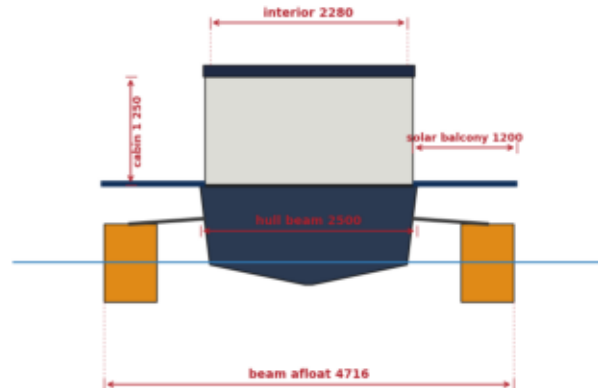


PROFILE — on the road (floats folded flat under the hull, towed stern-first)



6 × 205/70 R15 all-terrain · track 2270 · tongue load +200 kg · road width 2535 (limit 2 550)

MIDSHIP SECTION — afloat (looking forward)



KEY DIMENSIONS

OVERALL

length, hull
length with drawbar
beam — road / afloat
height on the road
air draft afloat
draft / displacement

LIVING QUARTERS

cabin length
cabin width, inside
standing headroom
roof sun deck
floor area
entry door, clear
cockpit floor / bulwark

DECKS & SOLAR

solar balcony
aft passage width
roof terrace
panels
5 laminates + 6 std modules (= 3.1 kWp)

ROAD GEAR

wheels
track / wheelbase
ground clearance
coupling height / tongue
float, each

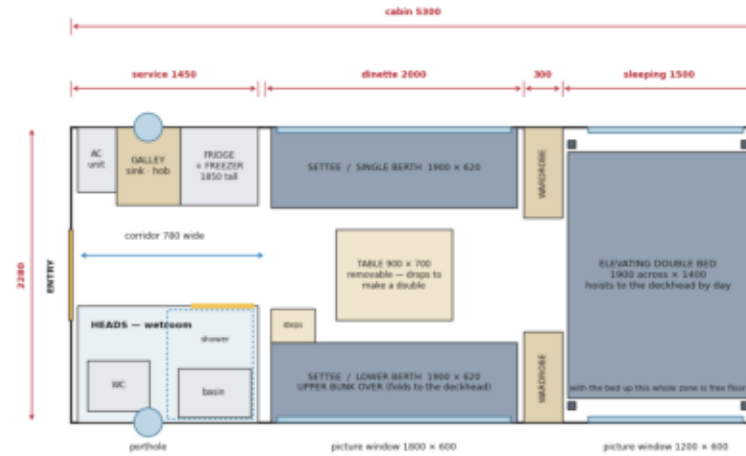
7 200 mm
9 094 mm
2 535 / 4 716 mm
3 009 mm (StVZO limit 4 000)
2 267 mm
260 mm / =1 970 kg
5 300 mm
2 280 mm
1 850 mm
walk-on glass, 8 panes
= 12.1 m²
700 × 1 700 mm
480 / 750 mm
1 200 wide: 0 walkway + 1160 panels
540 mm (part of the folding deck)
5 300 × 2 360 mm
5 laminates + 6 std modules (= 3.1 kWp)
6 × 205/70 R15 all-terrain
2 270 / 3 400 mm
482 mm
445 mm / +100 kg
6 200 × 600 × 900 mm

general arrangement

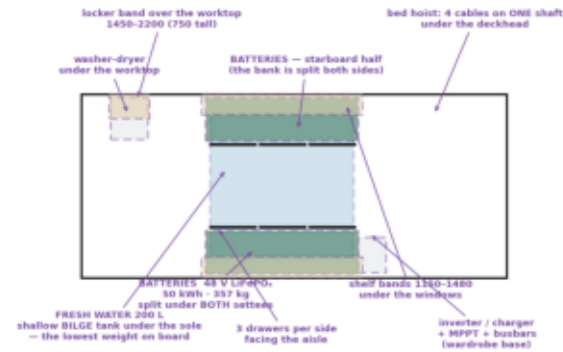
BOAT-HOME • INTERIOR

5300 x 2280 mm floor • 1850 mm headroom • berths for 4 • all dimensions in mm

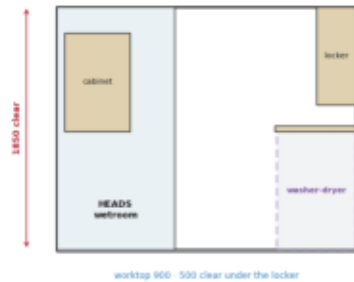
PLAN — what you see, bow to the right



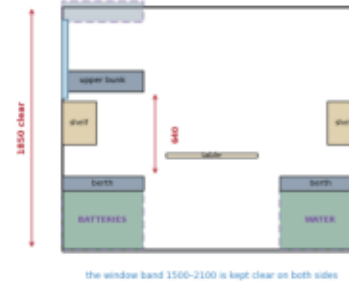
STOWAGE PLAN — what is hidden under and above it



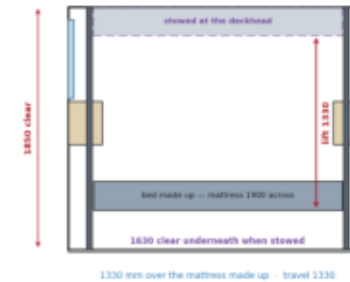
A-A galley / heads (looking forward)



B-B dinette (looking forward)



C-C elevating bed (looking forward)

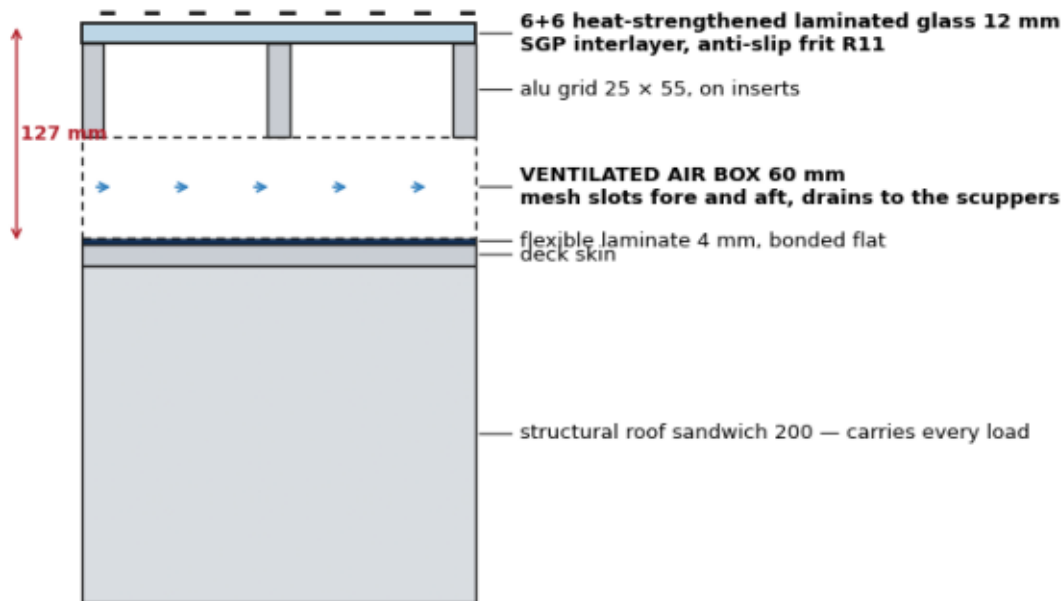


joinery berths / cushions appliances batteries 50 kWh / water — heavy, low hidden (dashed purple) glazing

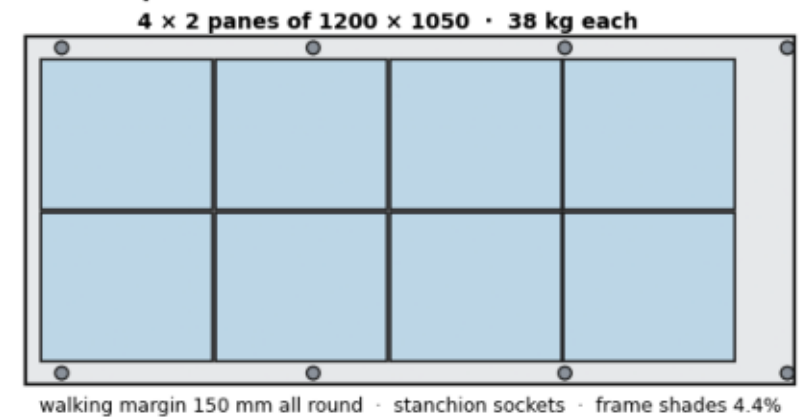
interior layout sheet

ROOF DECK BUILD-UP · walk on the glass, never on the panels

no moving parts anywhere on the roof; the air box ventilates the cells so they run cooler than bonded laminates



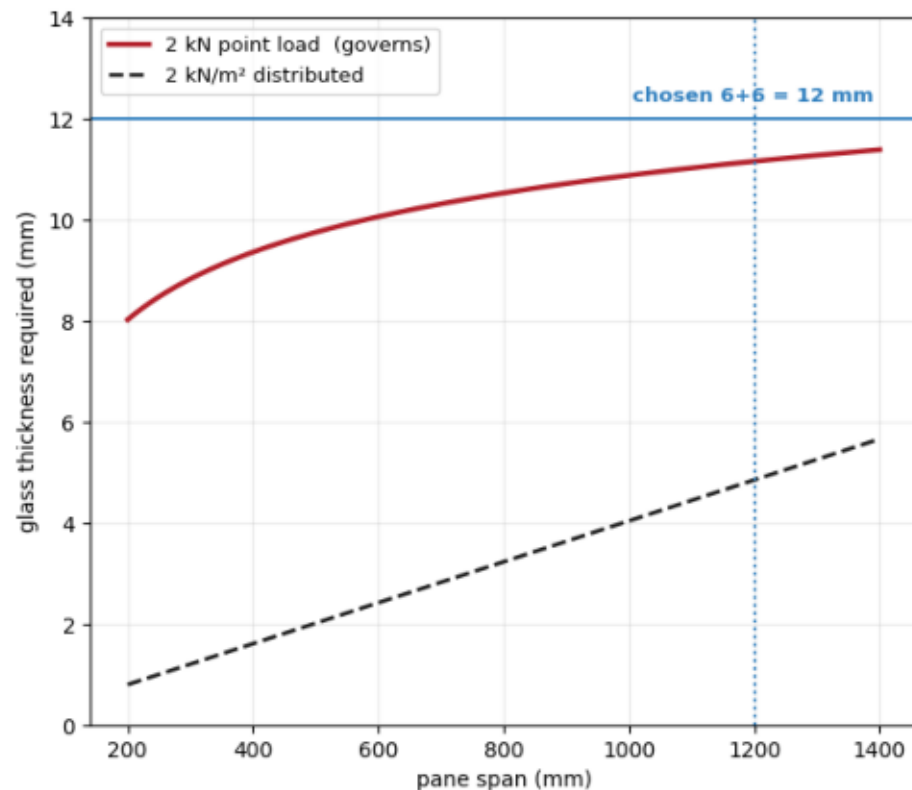
PLAN — 8 panes over the field



a cracked pane cannot drop anyone — only 60 mm of air under it,
so two plies is enough where a building floor needs three
(vertical scale x 6)

WALK-ON GLASS SIZING · why the pane size hardly matters

the concentrated load governs, and its stress grows with the LOG of the span — not the square of it



PANE SIZE TRADE (same 10 m² field)

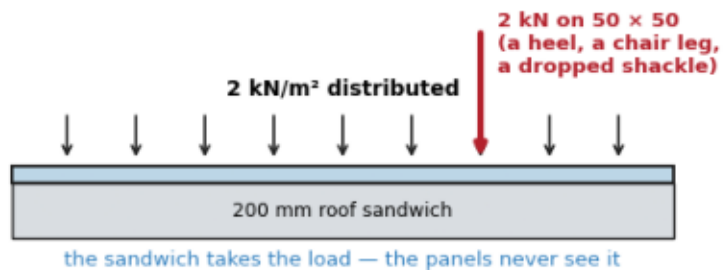
pane	glass	frame	TOTAL	shading
1000 mm → 12 mm	302kg	35kg	337kg	4.9%
500 mm → 10 mm	252kg	50kg	302kg	9.8%
400 mm → 10 mm	252kg	56kg	308kg	12.1%
200 mm → 10 mm	252kg	84kg	336kg	23.4%

CHOSEN

panes	4 × 2 of 1200 × 1050
glass	6+6 HS laminated, 12 mm, SGP
needed for point load	11.0 mm
deflection under UDL	0.9 mm (limit span/300 = 3.5)
mass per pane	38 kg
deck total	352 kg

ROOF DECK · loads, stability and what it generates

full building-code deck loads — no use restrictions, stilettos and dropped anchors included



STABILITY

4 crew hard to one side	3.7 kNm heeling
float righting moment	31.3 kNm
margin	8.5 ×

SOLAR

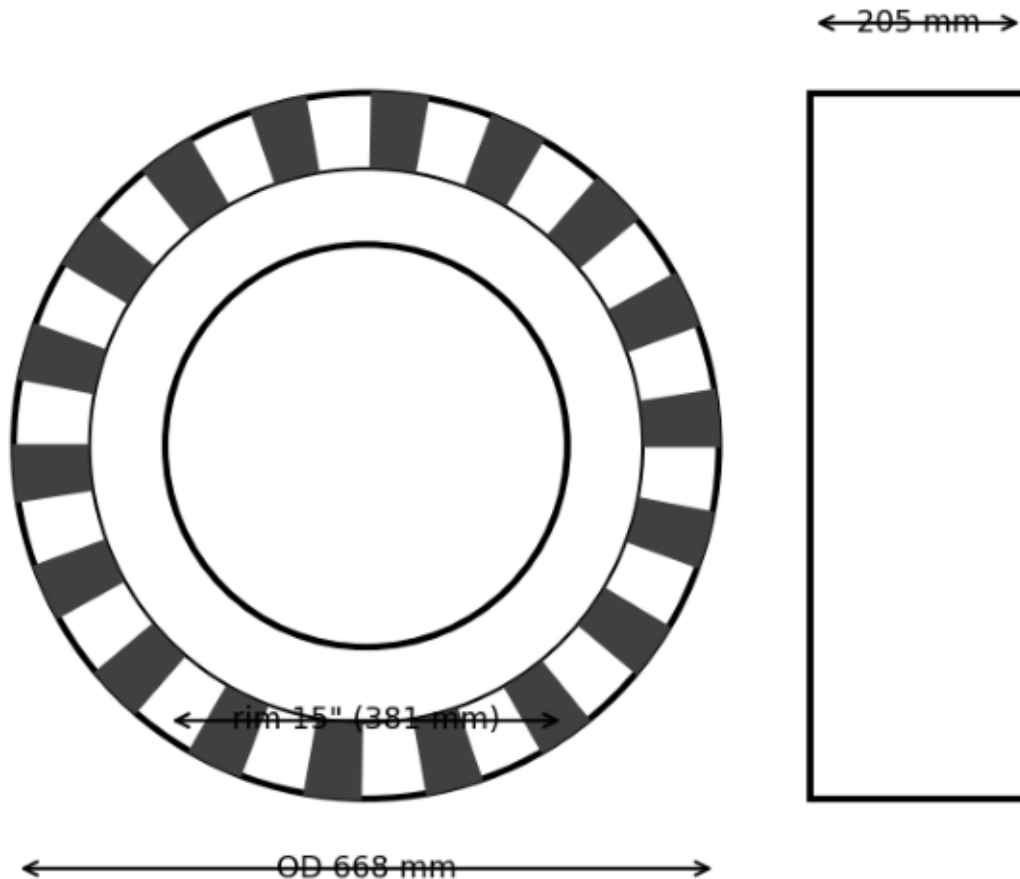
deck laminates	4 × 500 W = 2.00 kWp
solar balconies	2.00 kWp
nominal total	4.00 kWp
glass transmission	91 %
frame shading	-4.4 %
ventilation gain	+5 %
effective	3.83 kWp

GEOMETRY

deck surface	2527 above keel
air draft	2267 mm
road height	3009 mm (limit 4 000)
guardrail	1000 mm + 2 lifelines, removable

the deck generates best when nobody is on it — it is a sun deck first

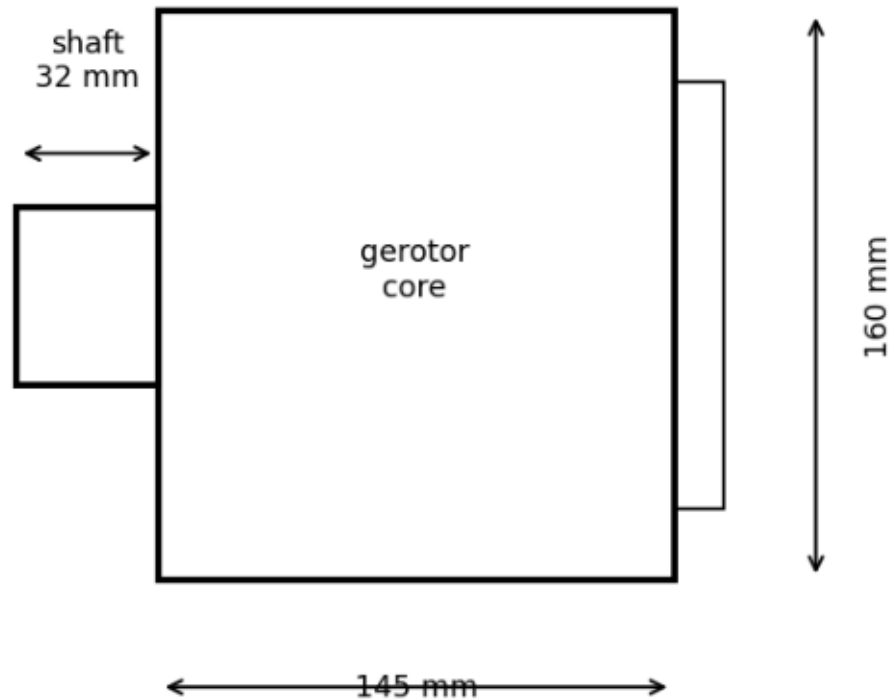
Tire — 205/70 R15 All-Terrain (Grabber AT3 class)



Specifications

Size: 205/70 R15
OD x width: 668 x 205 mm
Load index: 106 (950 kg) XL
Speed: T (190 km/h)
Tread: open-shoulder AT –
sand/mud traction, road-quiet
Pressure: 2.6 bar (road)
0.8-1.2 bar aired-down (sand)
Qty on boat: 6 (+1 spare)
Est. 2026: ~110 EUR/tire
Steel rim: ~45 EUR (15x6J ET30)

Hydraulic hub motor — orbital/gerotor, OMR-200 class



Specifications

Displacement: 200 cc/rev

Torque cont.: ~195 Nm (peak 320)

Speed range: 10-475 rpm
(wheel 84 rpm @ 10 km/h)

Max pressure: 140 bar cont.

Body: d160 x 145 mm, 6.5 kg

Shaft: d32 straight key

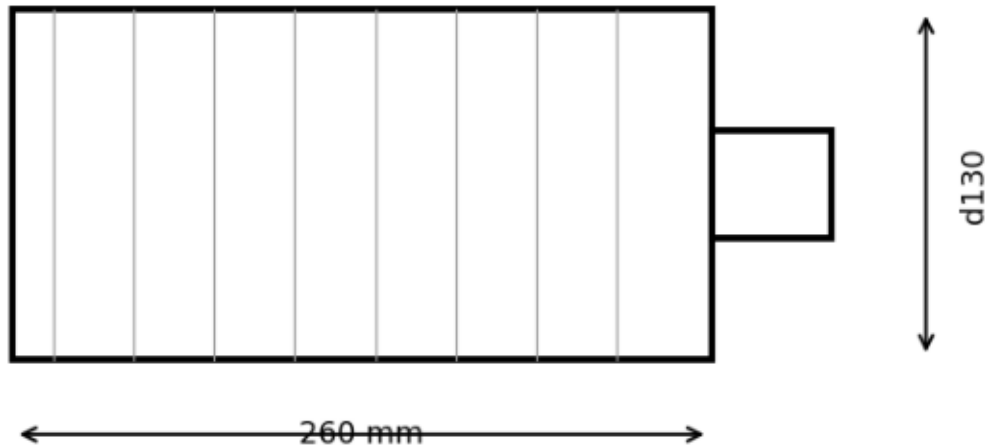
Mount: 4-bolt flange

Submersible: yes (oil-filled)

Qty: 6 Est. 2026: 180-260

EUR each (brand) / ~120 clone

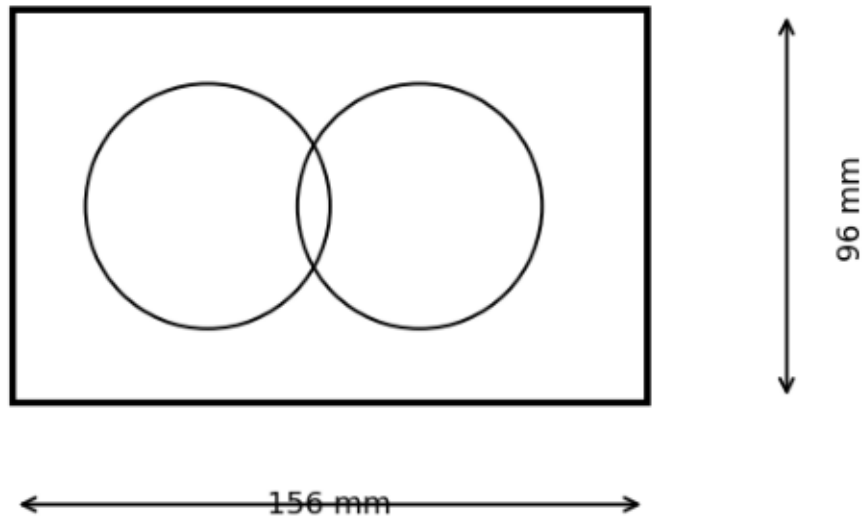
Bay motor — 48 V BLDC, 3 kW continuous



Specifications

Voltage: 48 V DC (house bank)
Power: 3 kW cont / 5 kW peak
Speed: ~3000 rpm to pump
Frame: d130 x 260 mm, ~9 kg
Cooling: oil-immersed in bay
Control: CAN/PWM controller
IP: 68 (potted, in bay)
Qty: 2 (one per float)
Est. 2026: ~350-450 EUR each
+ controller ~150 EUR

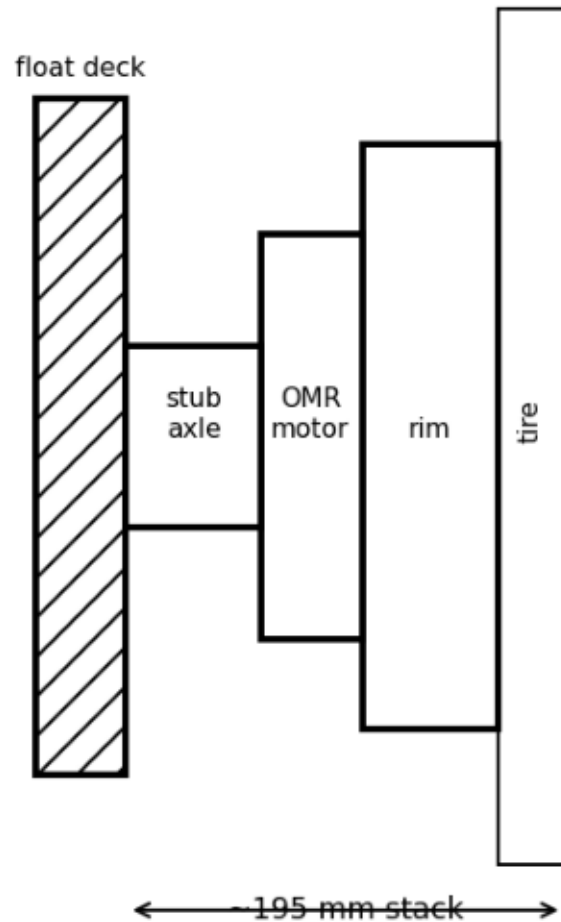
Hydraulic pump — Group 2 gear pump



Specifications

Displacement: 14 cc/rev
Flow: ~40 L/min @ 3000 rpm
Pressure: max 250 bar
(system runs $\leq$ 140 bar)
Body: 156 x 96 x 96 mm, 4 kg
Mount: EU flange, d80 spigot
Drive: direct on BLDC shaft
Qty: 2 Est. 2026: ~120-180
EUR each
+ valve manifold w/ prop. flow
divider: ~250-350 EUR/float

Hub stack — deck / stub axle / motor / wheel (section)



Specifications

Order (deck outward):

- 1 stub axle boss, welded
- 2 orbital motor, 4-bolt flange
- 3 wheel on motor hub (5x112)

Seals: labyrinth + 2x lip seal

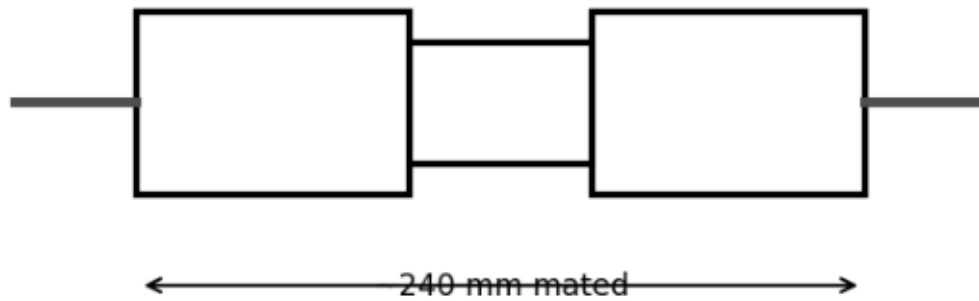
Anode: zinc ring inside cap

Cap: GRP clamshell, 3 latches

Service: wheel off = 5 bolts;

motor swap = 4 bolts + 2 hoses

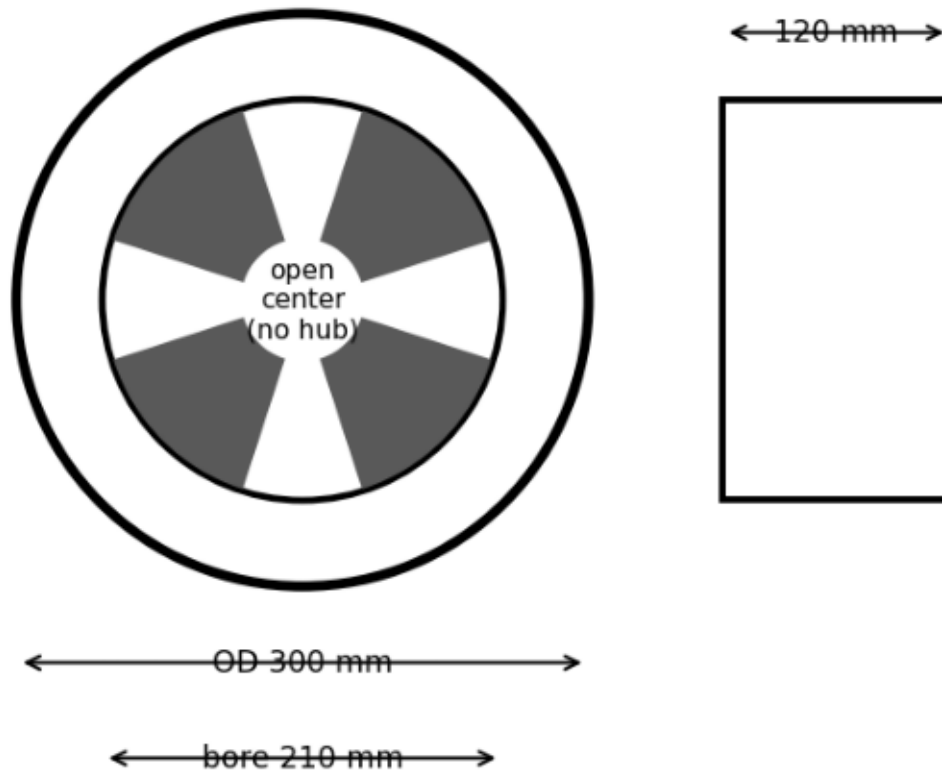
48 V feed — subsea wet-mate connector pair



Specifications

Type: 2P+PE wet-mateable
Rating: 60 V / 50 A, IP69K
(submerged mating allowed)
Cable: 6 mm² tinned, molded
Route: inside arm tubes,
shoulder + wrist joints only
Qty: 4 pairs Est. 2026: ~50-90
EUR/pair (industrial grade)

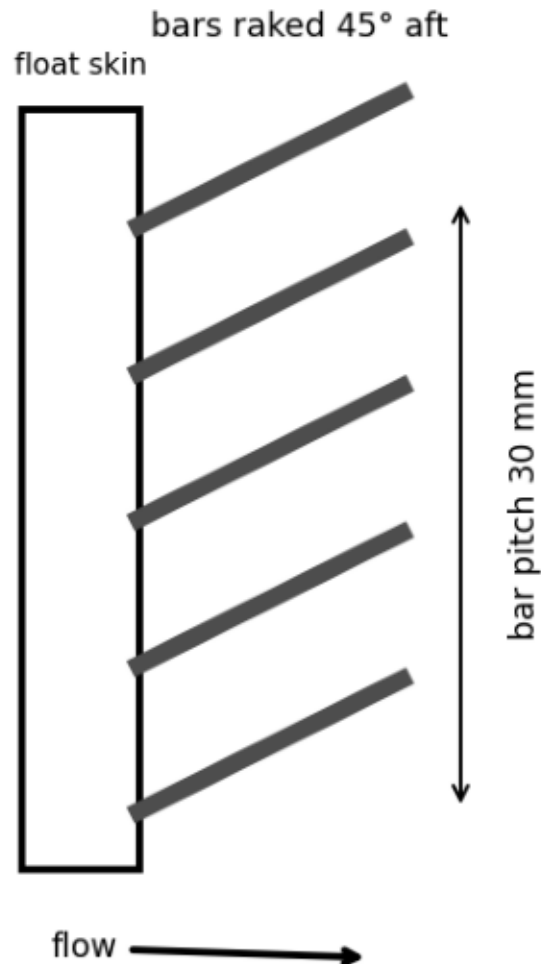
Rim-drive thruster — 48 V / 2 kW, hubless



Specifications

Type: rim-driven, flooded
stator, NO shaft, NO hub
Rotor: d200, 4 blades
Housing: OD 300 x 120 mm
Power: 2 000 W @ 48 V
Thrust: ~45-55 kgf static
Reverse: full (weed flush)
Submerged: native (wet motor)
Qty: 3 (+1 spare fits all)
Est. 2026: 2 500-4 000 EUR ea
(commercial marine unit)

Intake grille — raked trash-rack bars



Specifications

Bars: 5 x d9 stainless,
raked 45 deg aft
Pitch: ~30 mm
Loss: < 3 % intake area
Function: flow pushes weed
ALONG the bars, off the
trailing edge (trash-rack
practice); reverse burst
ejects anything caught
Both tunnel mouths, all 3 units
Est. 2026: workshop-made, ~40
EUR/set in 316 stainless